

SECTION 6.1

1. Label the following statements as true or false.

(a) An inner product is a scalar-valued function on the set of ordered pairs of vectors.

Answer : True. See the definition of an inner product.

(b) An inner product space must be over the field of real or complex numbers.

Answer : This book assumes only the inner product space over $\mathbb{R}$ or $\mathbb{C}$. However, we can also consider the inner product space $\mathbb{Q}$ over rational numbers $\mathbb{Q}$.

(c) An inner product is linear in both components.

Answer : False. Given a standard inner product, $\langle i, 1 \rangle = i$ but $\langle 1, i \rangle = -i$. To be precise,

$$\langle cx, y \rangle = c\langle x, y \rangle$$

and

$$\begin{aligned}\langle x, cy \rangle &= \overline{\langle cy, x \rangle} \\ &= \overline{c\langle y, x \rangle} \\ &= \bar{c}\langle x, y \rangle\end{aligned}$$

can be different.

(d) There is exactly one inner product on the vector space $\mathbb{R}^n$.

Answer : False. Let $u = (u_1, \dots, u_n)$ and $v = (v_1, \dots, v_n)$. Consider $\langle u \cdot v \rangle = 2 \sum_{i=1}^n u_i v_i$.

(e) The triangle inequality only holds in finite-dimensional inner product spaces.

Answer : False. Theorem 6.2 does not assume that the dimension of the inner product space is finite.

(f) Only square matrices have a conjugate-transpose.

Answer : False. This is also possible for any arbitrary matrix.

(g) If x, y , and z are vectors in an inner product space such that $\langle x, y \rangle = \langle x, z \rangle$, then $y = z$.

Answer : False. Let $x = (1, 0)$, $y = (0, 1)$ and $z = (0, 2)$. Given a standard inner product, we have $\langle x, y \rangle = \langle x, z \rangle = 0$. However, $y \neq z$.

(h) If $\langle x, y \rangle = 0$ for all x in an inner product space, then $y = 0$.

Answer : True. Since $\langle y, y \rangle = 0$, it follows that $y = 0$.

5. In $\mathbb{C}^2$, show that $\langle x, y \rangle = xAy^*$ is an inner product, where

$$A = \begin{pmatrix} 1 & i \\ -i & 2 \end{pmatrix}.$$

Compute $\langle x, y \rangle$ for $x = (1 - i, 2 + 3i)$ and $y = (2 + i, 3 - 2i)$.

Proof. First, we will show that $\langle x, y \rangle = xAy^*$ is an inner product.

$$(1) \quad \langle x + z, y \rangle = (x + z)Ay^* = xAy^* + zAy^* = \langle x, y \rangle + \langle z, y \rangle$$

$$(2) \quad \langle cx, y \rangle = cxAy^* = c(xAy^*) = c\langle x, y \rangle \text{ for any } c \in F$$

$$(3)$$

$$\begin{aligned} \overline{\langle x, y \rangle} &= \overline{xAy^*} \\ &= (\overline{xAy^*})^t \\ &= \overline{(xAy^*)^t} \\ &= y\overline{A^t}x^* \\ &= yAx^* \\ &= \langle y, x \rangle \end{aligned}$$

(4) Let $x = (a + bi, c + di) \in \mathbb{C}^2$. We need to show that $\langle x, x \rangle = xAx^*$. Since

$$\begin{aligned} xAx^* &= \begin{pmatrix} a + bi & c + di \end{pmatrix} \begin{pmatrix} 1 & i \\ -i & 2 \end{pmatrix} \begin{pmatrix} a - bi \\ c - di \end{pmatrix} \\ &= \begin{pmatrix} a + bi & c + di \end{pmatrix} \begin{pmatrix} a + d + (c - b)i \\ 2c - b + (-a - 2d)i \end{pmatrix} \\ &= a^2 + 2ad + 2d^2 + b^2 - 2bc + 2c^2 \\ &= (a + d)^2 + d^2 + (b - c)^2 + c^2 \end{aligned}$$

, it follows that $\langle x, x \rangle > 0$ if $x \neq 0$.

□

8. Provide reasons why each of the following is not an inner product on the given vector spaces.

(a) $\langle (a, b), (c, d) \rangle = ac - bd$ on $\mathbb{R}^2$.

Proof. This is not an inner product. Let $(a, b) = (1, 1)$. Then $\langle (a, b), (a, b) \rangle = a^2 - b^2 = 0$, but $(a, b) \neq (0, 0)$. $\square$

(b) $\langle A, B \rangle = \text{tr}(A + B)$ on $M_{2 \times 2}(\mathbb{R})$.

Proof. This is not an inner product. Let $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Then, we have

$$\langle A, A \rangle = \text{tr}(A + A) = \text{tr} \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} = 0.$$

However, the matrix $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \neq 0$. $\square$

(c) $\langle f(x), g(x) \rangle = \int_0^1 f'(t)g(t) dt$ on $P(\mathbb{R})$, where $'$ denotes differentiation.

Proof. This is not an inner product. Take $f(x) = 1$, constant functions. Then we have $\langle f(x), f(x) \rangle = 0$. $\square$

9. Let β be a basis for a finite-dimensional inner product space.

(a) Prove that if $\langle x, z \rangle = 0$ for all $z \in \beta$, then $x = 0$.

Proof. Since β is a basis for the finite-dimensional inner product space V , let $\beta = \{z_i : 1 \leq i \leq n\}$. For each $x \in V$, we have c_i such that

$$x = \sum_{i=1}^n c_i z_i.$$

Since $\langle x, z \rangle = 0$ for all $z \in \beta$, we have

$$\begin{aligned}
0 &= \sum_{i=1}^n \overline{c_i} \langle x, z_i \rangle \\
&= \sum_{i=1}^n \overline{c_i \langle z_i, x \rangle} \\
&= \sum_{i=1}^n \overline{c_i} \langle z_i, x \rangle \\
&= \sum_{i=1}^n \overline{\langle c_i z_i, x \rangle} \\
&= \sum_{i=1}^n \langle x, c_i z_i \rangle \\
&= \langle x, \sum_{i=1}^n c_i z_i \rangle \\
&= \langle x, x \rangle.
\end{aligned}$$

This implies that $x = 0$. □

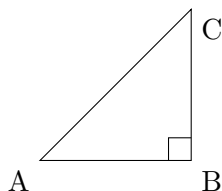
(b) Prove that if $\langle x, z \rangle = \langle y, z \rangle$ for all $z \in \beta$, then $x = y$.

Proof. We have $\langle x - y, z \rangle = 0$ for all $z \in \beta$. By (a), we have $x - y = 0$. Therefore, we conclude that $x = y$. □

10. Let V be an inner product space, and suppose that x and y are orthogonal vectors in V . Prove that $\|x + y\|^2 = \|x\|^2 + \|y\|^2$. Deduce the Pythagorean theorem in $\mathbb{R}^2$.

Proof. Note that $\langle x, y \rangle = \langle y, x \rangle = 0$ since x and y are orthogonal vectors in V . Thus, we have $\|x + y\|^2 = \langle x + y, x + y \rangle = \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle = \|x\|^2 + \|y\|^2$.

Consider a right triangle ABC with the right angle B .



Let $A = (0, 0)$ and $B = (x, 0)$ and $C = (x, y)$ in $\mathbb{R}^2$. Then,

$$|\overline{AC}|^2 = \|(x, 0) + (0, y)\|^2 = \|x\|^2 + \|y\|^2 = \overline{AB} + \overline{BC}.$$

□

11. Prove the parallelogram law on an inner product space V ; that is, show that

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2 \text{ for all } x, y \in V.$$

What does this equation state about parallelograms in $\mathbb{R}^2$?

Proof.

$$\begin{aligned} \|x + y\|^2 + \|x - y\|^2 &= \langle x + y, x + y \rangle + \langle x - y, x - y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle + \langle x, x \rangle - \langle x, y \rangle - \langle y, x \rangle + \langle y, y \rangle \\ &= 2\langle x, x \rangle + 2\langle y, y \rangle \\ &= 2\|x\|^2 + 2\|y\|^2 \end{aligned}$$

In $\mathbb{R}^2$, this equation implies that the sum of square of the four edges of a parallelogram is the sum of square of the two diagonals. □

13. Suppose that $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ are two inner products on a vector space V . Prove that $\langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle_1 + \langle \cdot, \cdot \rangle_2$ is another inner product on V .

Proof. (1)

$$\begin{aligned} \langle x + z, y \rangle &= \langle x + z, y \rangle_1 + \langle x + z, y \rangle_2 \\ &= \langle x, y \rangle_1 + \langle z, y \rangle_1 + \langle x, y \rangle_2 + \langle z, y \rangle_2 \\ &= \langle x, y \rangle_1 + \langle x, y \rangle_2 + \langle z, y \rangle_1 + \langle z, y \rangle_2 \\ &= \langle x, y \rangle + \langle z, y \rangle \end{aligned}$$

(2) For any $c \in F$,

$$\begin{aligned} \langle cx, y \rangle &= \langle cx, y \rangle_1 + \langle cx, y \rangle_2 \\ &= c\langle x, y \rangle_1 + c\langle x, y \rangle_2 \\ &= c(\langle x, y \rangle_1 + \langle x, y \rangle_2) \\ &= c\langle x, y \rangle \end{aligned}$$

(3)

$$\begin{aligned}
\overline{\langle x, y \rangle} &= \overline{\langle x, y \rangle_1 + \langle x, y \rangle_2} \\
&= \overline{\langle x, y \rangle_1} + \overline{\langle x, y \rangle_2} \\
&= \langle y, x \rangle_1 + \langle y, x \rangle_2 \\
&= \langle y, x \rangle
\end{aligned}$$

(4) If $x > 0$, then $\langle x, x \rangle = \langle x, x \rangle_1 + \langle x, x \rangle_2 > 0$ since $\langle x, x \rangle_1, \langle x, x \rangle_2 > 0$

□

15. (a) Prove that if V is an inner product space, then $|\langle x, y \rangle| = \|x\| \cdot \|y\|$ if and only if one of the vectors x or y is a multiple of the other. Hint: If the identity holds and $y \neq 0$, let

$$a = \frac{\langle x, y \rangle}{\|y\|^2},$$

and let $z = x - ay$. Prove that y and z are orthogonal and

$$|a| = \frac{\|x\|}{\|y\|}.$$

Then apply Exercise 10 to $\|x\|^2 = \|ay + z\|^2$ to obtain $\|z\| = 0$.

Proof. First, suppose that $|\langle x, y \rangle| = \|x\| \cdot \|y\|$. If $y = 0$, then $y = 0(x)$. This shows that y is a multiple of x . Now, we assume that $y \neq 0$. Let

$$a = \frac{\langle x, y \rangle}{\|y\|^2},$$

and let $z = x - ay$. Since

$$\begin{aligned}
\langle z, y \rangle &= \langle x - ay, y \rangle \\
&= \langle x, y \rangle - a\langle y, y \rangle \\
&= \langle x, y \rangle - \langle x, y \rangle \\
&= 0
\end{aligned}$$

, z and y are orthogonal. Also, we have

$$(0.1) \quad |a| = \frac{|\langle x, y \rangle|}{\|y\|^2} = \frac{\|x\| \cdot \|y\|}{\|y\|^2} = \frac{\|x\|}{\|y\|}.$$

Applying Exercise 10 to $\|x\|^2 = \|ay + z\|^2$, we have $\|x\|^2 = \|ay\|^2 + \|z\|^2$. By equality (??), we have $\|x\|^2 = \|ay\|^2$. Hence, $\|z\|^2 = 0$. This implies that $x = ay$.

Conversely, we assume that $x = ay$ for some $a \in F$. Then it is easy to see that $|\langle x, y \rangle| = \|x\| \cdot \|y\|$.

□

(b) Derive a similar result for the equality $\|x + y\| = \|x\| + \|y\|$, and generalize it to the case of n vectors.

Proof. First, assume that $x = cy$ for some $c \in F$. Then,

$$\begin{aligned} \|x + y\| &= \|(c + 1)y\| \\ &= (c + 1)\|y\| \\ &= c\|y\| + \|y\| \\ &= \|cy\| + \|y\| \\ &= \|x\| + \|y\|. \end{aligned}$$

Conversely, assume that $\|x + y\| = \|x\| + \|y\|$. Then we have $|\langle x, y \rangle| = \|x\| \cdot \|y\|$. (cf. the proof of Theorem 6.2(d)). By exercise 15. (a), we conclude that x or y is a multiple of the other.

Now, we want to generalize this result to the case of n vectors $x_1, \dots, x_n$. If $n - 1$ vectors are multiples of another vector, then it is easy to see that

$$\|x_1 + \dots + x_n\| = \|x_1\| + \dots + \|x_n\|$$

(see the previous case with two vectors).

We assume that

$$\|x_1 + \dots + x_n\| = \|x_1\| + \dots + \|x_n\|.$$

By Theorem 6.2(d), we have

$$\begin{aligned} \|x_1\| + \dots + \|x_n\| &= \|x_1 + \dots + x_n\| \\ &\leq \|x_1 + \dots + x_{n-1}\| + \|x_n\| \\ &\leq \|x_1 + \dots + x_{n-2}\| + \|x_{n-1}\| + \|x_n\| \\ &\vdots \\ &= \|x_1 + x_2\| + \|x_3\| + \dots + \|x_n\|. \end{aligned}$$

This implies that x_1 or x_2 is a multiple of the other. Thus, if we choose any x_i and x_j ($i \neq j$), then we can see that x_i or x_j is a multiple of the other. Therefore, we conclude that $n - 1$ vectors are multiples of another vector.

□

20. Let V be an inner product space over F . Prove the polar identities: For all $x, y \in V$,

$$(a) \langle x, y \rangle = \frac{1}{4} \|x + y\|^2 - \frac{1}{4} \|x - y\|^2 \text{ if } F = \mathbb{R};$$

Proof.

$$\begin{aligned} \frac{1}{4} \|x + y\|^2 - \frac{1}{4} \|x - y\|^2 &= \frac{\langle x, x \rangle + 2\langle x, y \rangle + \langle y, y \rangle}{4} - \frac{\langle x, x \rangle - 2\langle x, y \rangle + \langle y, y \rangle}{4} \\ &= \langle x, y \rangle \end{aligned}$$

□

$$(b) \langle x, y \rangle = \frac{1}{4} \sum_{k=1}^4 i^k \|x + i^k y\|^2 \text{ if } F = \mathbb{C}, \text{ where } i^2 = -1.$$

Proof.

$$\begin{aligned} \frac{1}{4} \sum_{k=1}^4 i^k \|x + i^k y\|^2 &= \frac{1}{4} (i \|x + iy\|^2 - \|x - y\|^2 - i \|x - iy\|^2 + \|x + y\|^2) \\ &= \frac{1}{4} (i(\|x + iy\|^2 - \|x - iy\|^2) + (\|x + y\|^2 - \|x - y\|^2)) \\ &= \frac{1}{4} (2i(\langle x, iy \rangle + \langle iy, x \rangle) + 2(\langle x, y \rangle + \langle y, x \rangle)) \\ &= \frac{1}{4} (-2i^2 \langle x, y \rangle + 2i^2 \langle y, x \rangle) + 2(\langle x, y \rangle + \langle y, x \rangle) \\ &= \frac{1}{4} (4\langle x, y \rangle) \\ &= \langle x, y \rangle \end{aligned}$$

□